

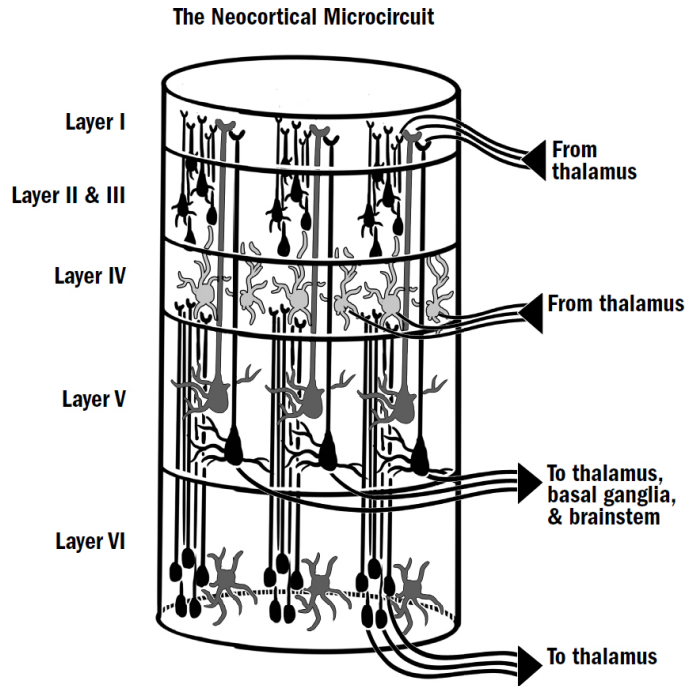


Figure 11.3: The microcircuitry of the neocortical column

Original art by Mesa Schumacher

The question is, of course: What is the computation?

Peculiar Properties of Perception

In the nineteenth century, the scientific study of human perception began in full force. Scientists around the world started probing the mind. How does vision work? How does audition work?

The inquiry into perception began with the use of illusions; by manipulating people's visual perceptions, scientists uncovered three peculiar properties of perception. And because much of perception, in humans at least, occurs in the neocortex, these properties of perception teach us about how the neocortex works.

Property #1: Filling In

The first thing that became clear to these nineteenth-century scientists was that the human mind automatically and unconsciously fills in missing things. Consider the images in [figure 11.4](#). You immediately perceive the word *editor*. But this is not what your eye is actually seeing—most of the lines of the letters are missing. In the other images too, your mind perceives something that is not there: a triangle, a sphere, and a bar with something wrapped around it.

This filling in is not a property unique to vision; it is seen across most of our sensory modalities. This is how you can still understand what someone is saying through a garbled phone connection and how you can identify an object through touch even with your eyes closed.



Figure 11.4: Filling-in property of perception

“Editor” from Jastrow, 1899. Others from Lehar, 2003.

Property #2: One at a Time

If your mind fills in what it thinks is there based on sensory evidence, what happens if there are multiple ways to fill in what you see? All three of the images in [figure 11.5](#) are examples of visual illusions devised in the 1800s to probe this question. Each of these images can be interpreted in two different ways. On the left side of [figure 11.5](#), you can see it as a staircase, but you can also see it as a protrusion from *under* a staircase.* In the middle of [figure 11.5](#), the cube could be one where the bottom right square is the front or where the top left square is the front. On the right of [figure 11.5](#), the picture could be a rabbit or a duck.

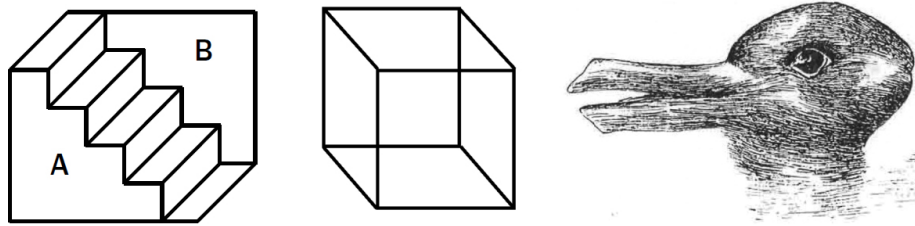


Figure 11.5: One-at-a-time property of perception

Staircase from Schroeder, 1858. “Necker cube” from Necker, 1832. Duck or rabbit from from Jastrow, 1899.

What is interesting about all these ambiguous pictures is that your brain can see only one interpretation at a time. You cannot see a duck and a rabbit simultaneously, even though the sensory evidence is equally suggestive of both. The mechanisms of perception in the brain, for some reason, require it to pick only one.

This also applies to audition. Consider the “cocktail-party effect.” If you are at a noisy cocktail party, you can tune in to the conversation of the person you are speaking to or the conversation of a nearby group. But you cannot listen to both conversations at the same time. No matter which conversation you tune in to, the auditory input into your ear is identical; the only difference is what your brain infers from that input. You can perceive only a single conversation at a time.

Property #3: Can’t Unsee

What happens when sensory evidence is *vague*—when it isn’t clear that it can be interpreted as anything meaningful at all? Consider the image in 11.6. If you haven’t seen these before, they will look like nothing—just blobs. If I give you a reasonable interpretation of these blobs, all of a sudden, your perception of them will change.

Figure 11.6 can be interpreted as a frog (see [here](#) if you don’t see this). Once your mind perceives this interpretation, you will never be able to unsee it. This is what might be called the can’t-unsee property of perception. Your mind likes to have an interpretation that explains sensory input. Once I give you a good explanation, your mind sticks to it. You now perceive a frog.



Figure 11.6: The can't-unsee property of perception

Image from Fahle et al., 2002. Used with permission by The MIT Press.

In the nineteenth century, a German physicist and physician named Hermann von Helmholtz proposed a novel theory to explain these properties of perception. He suggested that a person doesn't perceive what is experienced; instead, he or she perceives what the brain *thinks* is there—a process Helmholtz called *inference*. Put another way: you don't perceive what you actually see, you perceive a simulated reality that you have *inferred* from what you see.

This idea explains all three of these peculiar properties of perception. Your brain fills in missing parts of objects because it is trying to decipher the truth that your vision is *suggesting* (“Is there actually a sphere there?”). You can see only one thing at a time because your brain must pick a single reality to simulate—in reality, the animal can't be both a rabbit and a duck. And once you see that an image is best explained as a frog, your brain maintains this reality when observing it.

While many psychologists came to agree, in principle, with Helmholtz's theory, it would take another century before anyone proposed how Helmholtz's perception by inference might actually work.

Generative Models: Recognizing by Simulating